

ZHURAVLEVKA, Russian Federation

WMO: 319420

Lat: **44.750N**Lon: **134.467E**Elev: **170**StdP: **99.30**Time Zone: **10.00 (E10)**Period: **82-97**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.5	-33.2	-38.0	0.1	-35.5	-35.5	0.1	-33.1	6.3	-16.4	5.5	-15.7	0.1	50	0.521

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.5	30.5	22.0	28.6	20.9	27.0	20.1	23.7	28.7	22.6	26.7	21.6	25.2	1.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.2	26.1	21.2	16.2	25.2	20.3	15.2	24.0	72.2	28.9	67.8	26.7	63.9	25.3	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.2	5.3	4.3	-39.2	33.5	3.2	1.9	-41.6	34.9	-43.5	36.0	-45.3	37.1	-47.7	38.5		
(5)			-39.0	25.9	3.1	1.9	-41.2	27.3	-43.0	28.4	-44.7	29.5	-47.0	30.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.3	-21.3	-16.0	-6.3	5.3	11.9	16.9	20.9	20.9	13.6	4.8	-6.9	-17.6
	DBStd	15.34	4.89	5.53	5.50	3.90	3.63	2.98	2.58	2.85	3.62	4.37	6.01	5.97
	HDD10.0	3910	969	727	506	148	22	0	0	0	11	167	506	856
	HDD18.3	6052	1227	960	764	391	199	63	6	7	146	419	756	1114
	CDD10.0	1097	0	0	0	7	82	206	339	337	119	7	0	0
	CDD18.3	196	0	0	0	0	2	18	87	85	4	0	0	0
	CDH23.3	1637	0	0	0	8	78	250	648	624	27	2	0	0
	CDH26.7	431	0	0	0	1	12	53	184	178	2	0	0	0
Wind	WSAvg	1.2	0.8	1.0	1.4	1.8	1.7	1.2	1.1	1.0	1.0	1.4	1.3	0.8
Precipitation	PrecAvg	845	27	28	47	44	80	84	125	132	97	73	62	48
	PrecMax	1101	65	73	95	103	143	143	291	273	278	149	105	101
	PrecMin	505	3	2	4	8	9	37	51	29	20	21	18	12
	PrecStd	136	15	21	24	22	30	30	46	57	58	32	25	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.8	2.8	12.4	23.8	28.2	30.3	32.9	33.3	25.9	21.1	10.7	1.8
		MCWB	-4.9	-0.1	5.2	11.2	15.8	20.0	21.8	23.3	19.2	12.6	5.3	0.9
	2%	DB	-7.8	-0.5	7.3	20.0	25.5	28.4	30.5	30.7	23.6	18.0	6.7	-2.1
		MCWB	-9.1	-2.5	2.2	9.9	14.3	19.1	22.2	23.3	17.6	11.3	3.0	-3.3
	5%	DB	-10.4	-3.2	5.1	17.0	22.9	26.4	28.5	28.4	21.8	15.3	4.0	-4.8
		MCWB	-11.4	-5.0	0.7	8.1	13.0	18.3	21.3	22.2	16.4	9.3	1.0	-5.8
	10%	DB	-12.6	-5.9	2.9	14.1	20.4	24.4	26.9	26.7	20.2	13.0	1.6	-7.6
		MCWB	-13.5	-7.6	-0.6	6.8	12.1	17.5	20.6	21.1	15.7	8.2	-0.7	-8.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.2	0.1	5.5	12.1	17.4	23.1	25.1	25.1	21.3	14.0	6.7	0.9
		MCDB	-3.4	2.1	11.4	22.0	24.9	27.2	29.7	30.3	24.2	20.0	9.9	1.4
	2%	WB	-9.0	-2.4	2.6	10.3	15.5	20.4	23.6	24.0	19.0	11.9	3.6	-3.2
		MCDB	-7.9	-0.6	6.7	18.7	22.1	26.3	28.0	29.5	21.9	16.4	6.2	-2.2
	5%	WB	-11.4	-4.9	1.0	8.6	14.1	19.0	22.5	23.0	17.5	10.3	1.4	-5.8
		MCDB	-10.5	-3.1	4.4	15.8	20.8	24.9	26.5	27.5	20.7	14.4	3.4	-4.9
	10%	WB	-13.5	-7.5	-0.3	7.1	13.0	17.8	21.5	22.0	16.3	8.6	-0.4	-8.6
		MCDB	-12.6	-5.9	2.5	13.4	18.6	23.2	25.3	25.7	19.4	12.2	1.4	-7.7
Mean Daily Temperature Range	5% DB	MDBR	15.3	17.2	14.6	13.6	13.5	11.9	10.5	10.2	11.7	13.1	12.2	13.6
		MCDBR	15.9	18.4	16.4	20.7	19.2	16.6	13.9	12.7	14.3	18.2	14.5	12.5
		MCWBR	14.7	15.7	11.0	11.0	9.0	8.2	6.5	6.1	8.1	10.9	10.7	11.4
	5% WB	MCDBR	15.7	16.8	14.1	18.8	15.8	14.7	11.1	12.1	12.0	16.0	13.5	12.2
		MCWBR	14.5	14.4	9.8	10.4	7.8	7.7	6.1	6.3	7.4	10.0	10.4	11.2
Clear-Sky Solar Irradiance	taub	0.308	0.367	0.442	0.540	0.549	0.529	0.488	0.456	0.414	0.422	0.373	0.328	
	taud	2.090	1.915	1.802	1.743	1.802	1.946	2.128	2.222	2.253	2.074	2.092	2.054	
	Ebn at Noon	774	778	767	728	739	758	786	799	801	716	682	693	
	Edh at Noon	99	144	186	215	209	182	150	132	118	123	100	94	
All-Sky Solar Radiation	RadAvg	2.14	3.14	4.07	4.52	5.02	5.38	4.93	4.53	3.99	2.84	1.93	1.66	
	RadStd	0.14	0.17	0.30	0.43	0.44	0.59	0.48	0.37	0.34	0.30	0.20	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page